

Namrata Yadav

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EDUCATION

MS in Computer Engineering, San Diego State University, CA (GPA: 3.4/4.0)

Aug 2024 - Dec 2026

B.Tech. in Electrical & Electronics Engineering, CVR College of Engineering, India (GPA: 3.5/4.0)

Aug 2019 - May 2023

TECHNICAL SKILLS

EDA Tools: Cadence Virtuoso, Layout XL, Spectre, ADE Explorer/Assembler, Maestro, Assura; Synopsys Custom Compiler, IC Validator, StarRC, HSPICE, PrimeWave, Design Compiler, IC Compiler II, PrimeTime; Siemens Calibre; Magic, Netgen, Xschem, ngspice, KLayout

PDKs & Technologies: ASAP7 7 nm FinFET, SAED 28/32 nm, GPD45, SKY130, 0.35 μ m CMOS

Layout & Physical Verification: Full-custom analog, mixed-signal and digital layout; logic gates, flip-flops, transmission gates, standard cells and SRAM bitcells; device matching, parasitic- and EM/IR-aware routing; DRC, LVS, ERC, antenna, density/fill, PEX

Analog, Mixed-Signal & Reliability: Op-amps/OTAs, bandgap references, LDOs, CTAT/PTAT sensors; gm/ID sizing, compensation and stability; device characterization, VT/gm/ID extraction; DC/AC/transient analysis, PSRR, noise, PVT corners, Monte Carlo mismatch, MOSRA/URI reliability aging and lifetime drift

Digital Design & Physical Design: Verilog RTL design and testbench verification; RTL-to-GDSII flow, logic synthesis, floorplanning, power planning, placement, clock tree synthesis, routing, SDC, multi-corner static timing analysis, setup and hold timing closure, PPA

Programming & Environment: Python, Tcl; Linux/UNIX, Bash/C-shell scripting, EDA log inspection, PDK/library setup, Git

EXPERIENCE

Research Assistant, Neuromorphic Hardware Co-Design | SDSU

Nov 2025 - Present

- Designed a CMOS leaky integrate-and-fire neuron (0.35 μ m, 3.2 V) with weighted current-mirror synapses, a subthreshold leak path, inverter comparator and reset feedback, sizing mirror ratios 1x/5x/8x for a near-linear 8.75 Hz/nA F-I response and 4.86 ms membrane time constant.
- Built the full-custom layout of the neuron with a MOSCAP membrane array (42 μ m/800 nm, m=10), holding mirror matching through common-centroid placement and identical device orientation and routing the high-impedance membrane node on lower metal clear of switching signals; closed DRC and LVS and validated behavior in post-layout simulation.
- Extended the cell into a clock-gated two-neuron network with a CLK-switched current-mirror input stage and an 8x output mirror fired by the upstream spike, characterizing propagation across a 1-15 nA DC input sweep; correlated Cadence spike trains against a Python/SpikingJelly model reaching 94.5% MNIST accuracy versus 97.4% ideal, and is scaling the same flow to a 20-to-1 array.

Graduate Teaching Associate | SDSU

Aug 2025 - Present

Lead circuits, electromagnetism and optics labs for ~50 students per semester: instrumentation, measurement methodology, debugging.

IC Layout Design Trainee, Takshila Institute of VLSI Technologies | India

Aug 2023 - Apr 2024

- Built full-custom CMOS and FinFET layouts for combinational logic cells and a five-stage ring oscillator, applying device matching, diffusion sharing and fin-quantized sizing; debugged spacing, enclosure, connectivity and device-geometry violations.

PROJECTS

Analog and Mixed-Signal IC Layout

Cadence Virtuoso, Layout XL, GPD45

- Developed custom layouts for 50+ analog and mixed-signal blocks (op-amp, bandgap reference, LDO, level shifter, Schmitt trigger, PLL), building the core as a 5 x 5 common-centroid BJT array with a dummy ring and interdigitating differential pairs and mirrors elsewhere.
- Built the full-custom layout of a 1.8 V to 1.2 V LDO (5T OTA, thick-oxide PMOS pass device), routing the 96 μ m multi-finger pass device with wide metal and multi-via rails to spread current across fingers for EM and IR-drop headroom; traced post-layout shifts through extraction and lifetime drift with MOSRA/URI aging, measuring 1.192 V output at 149.8 mV dropout.

7 nm FinFET Standard-Cell Design and Characterization

ASAP7, Cadence Virtuoso, Calibre, HSPICE

- Designed and built the full-custom layout of a NAND2xp67 standard cell in ASAP7 7 nm FinFET technology, verifying functionality at 0.7 V and achieved DRC/LVS-clean layout, and performed 3 x 3 cell-abutment verification.
- Ran PEX and pre/post-layout timing characterization in HSPICE, quantifying extracted interconnect R/C impact on propagation delay and rise/fall times.

28/32 nm CMOS Standard-Cell Library and SRAM Layout

Synopsys Custom Compiler, IC Validator, StarRC

- Built an 11-track, eight-cell standard-cell library (INV, BUF, NAND2, NOR2, AND2, OR2, XOR2, XNOR2) on a fixed poly pitch, uniform cell height, aligned VDD/VSS rails and grid-aligned M1 pin access, using Euler-path ordering for diffusion sharing; composed them into muxes, flip-flops, counters and an 8:1/1:8 SerDes, confirming rail alignment and abutment across assembly.
- Built transistor-level layout for a 6T SRAM bitcell and array with shared rails, word-line distribution, symmetric bit-lines for sense margin and well taps for latch-up immunity; extracted parasitics in StarRC and verified timing in PrimeWave.

CMOS Temperature Sensor: CTAT Design and Device Characterization

Cadence Virtuoso, Spectre, Maestro, GPD45

- Designed a CTAT/PTAT sensor around a Miller-compensated op-amp, characterized -25 C to 125 C, and extracted VT/gm/ID trends to select a 2 μ A CTAT bias point monotonic at -0.201 mV/C, confirmed under post-layout PVT, Monte Carlo and aging.

SKY130 Physical Verification and LVS Debugging

Magic, Netgen, Xschem, ngspice, KLayout

- Took standard-cell and analog blocks to DRC- and LVS-clean in the open-source SKY130 flow, running DRC and parasitic extraction in Magic, netlisting schematics in Xschem and batch LVS in Netgen; scripted the PDK paths and shell setup for repeatable runs.
- Resolved connectivity, port, black-box, hierarchical and device-property mismatches down to the offending net or instance, and inspected GDS in KLayout.

32-bit Pipelined ALU: RTL to GDSII and Timing Closure

Synopsys SAED 32 nm

- Designed a 32-bit three-stage pipelined ALU in Verilog and verified it against an independent reference model across 31,691 randomized transactions with zero mismatches, covering arithmetic, logic, shift and pipeline hazard paths.
- Automated the RTL-to-GDSII flow in Tcl and Python through synthesis, floorplanning, power planning, placement, CTS and routing, reporting multi-corner STA at each stage.